

Assignment - 02

1) What are the different protocols you observe at the following layers of the protocol stack?

a) Application layer

- HTTP - Used when accessing `http://neverssl.com`.
- HTTPS - Secure web communication for `https://www.Cornell.edu`.
- TLS - Provides encryption for HTTPS connections.
- DNS - Resolves domain names into IP addresses.

b) Transport layer

- TCP - Carries HTTP, HTTPS, and TLS traffic.
- UDP - Commonly used for DNS queries (unless DNS falls back to TCP).

c) Network layer

- IPv4 - Most commonly observed protocol for routing packets.
- IPv6 - May also appear if your system/network supports IPv6.
- ICMP - May appear for network diagnostics (not always present).

2) What is the total amount of data being received for the following two cases?

Ans → website

<http://neverssl.com/>

Total Data Received
3,659 bytes (≈ 3.57 KB)

<https://www.cornell.edu/>

5,579,820 bytes
(≈ 5.32 MB)

3) Answer the following when you access the site <http://neverssl.com/>.

a) How many HTTP Get request do you observe?
list down the Get requests.

Ans → There are HTTP Get requests.

- Get /online HTTP/1.1
- GET /Online HTTP/1.1
- Get /Favicon.ico

b) What version of HTTP the server is running?

Ans → HTTP/1.1

c) For each of the HTTP Get request find out.

i) The total no. of TCP segments being received.

Ans → 8,638 packets.

ii) The total amount of data being received in the corresponding HTTP response message.

Ans → HTTP/1.1 200K

Content-length = 241

D) Identify TCP three way handshake used to establish the connection with the <http://neverssl.com>. Specify SYN, SYN-ACK, ACK packets and write the sequence no. of each.

Ans

Packet

SYN

→

SYN-ACK

→

ACK

→

Sequence Number.

23.64.59.64

8.219.181.32

10.85.18.185